THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of prime factors (counted with repetition) of 72 is: 1

(a) 12 **(b)** 4 **(c)** 5 **(d)** 6

2. Which of the following is an irrational number? 1

(a) $\sqrt{16}$ **(b)** $\sqrt{15}$ **(c)** 16 **(d)** $\frac{4}{7}$

3. The prime factorisation of 126 is: 1

(a) $2^2 \times 31$ **(b)** $2 \times 3^2 \times 7$ **(c)** $2 \times 5 \times 13$ **(d)** 2^7

4. The decimal expansion of $\frac{5}{12}$ is: 1

(a) non-terminating non-repeating **(b)** non-terminating repeating

(c) an integer **(d)** terminating

5. The HCF of 28 and 42 is: 1

(a) 14 **(b)** 7 **(c)** 84 **(d)** 28

6. The LCM of 12 and 16 is: 1

(a) 4 **(b)** 28 **(c)** 48 **(d)** 192

7. The number of prime factors (counted with repetition) of 90 is: 1

(a) 12 **(b)** 5 **(c)** 4 **(d)** 3

8. Which of the following is an irrational number? 1

(a) $\frac{8}{11}$ **(b)** $\sqrt{64}$ **(c)** $\sqrt{17}$ **(d)** 64

9. The HCF of 16 and 24 is: 1

(a) 48 **(b)** 4 **(c)** 16 **(d)** 8

10. The LCM of 14 and 21 is: 1

(a) 42 **(b)** 7 **(c)** 294 **(d)** 35

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Without long division, state whether $\frac{17}{125}$ is terminating or non-terminating. 2

12. Find the HCF of 42 and 70 by prime factorisation. 2

13. Express 504 as a product of prime factors. 2

14. Without long division, state whether $\frac{19}{21}$ is terminating or non-terminating. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the LCM and HCF of 45 and 75 by prime factorisation. 3

16. Verify that HCF $\times$ LCM $=$ product of the numbers for 24 and 40. 3

17. Express $\frac{5}{6}$ as a decimal after checking its type. 3

OR

Find the LCM and HCF of 54 and 90 by prime factorisation. 3

18. Verify that HCF $\times$ LCM $=$ product of the numbers for 28 and 70. 3

OR

Express $\frac{4}{9}$ as a decimal after checking its type. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the HCF and LCM of 306 and 408 by prime factorisation and verify that HCF $\times$ LCM $=$ product. 5

OR

Find the LCM and HCF of 18, 27 and 36 using prime factorisation. 5

20. Find the HCF and LCM of 420 and 588 by prime factorisation and verify that HCF $\times$ LCM $=$ product. 5

OR

Find the LCM and HCF of 24, 36 and 60 using prime factorisation. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND